



Published in final edited form as:

Annu Rev Biophys. 2024 July ; 53(1): 87–108. doi:10.1146/annurev-biophys-030722-020555.

The Effects of Codon Usage on Protein Structure and Folding

McKenze J. Moss,

Laura M. Chamness,

Patricia L. Clark*

Department of Chemistry & Biochemistry, University of Notre Dame, Notre Dame, IN 46556 USA

Abstract

The rate of protein synthesis is slower than many folding reactions and varies depending on the synonymous codons encoding the protein sequence. Synonymous codon substitutions thus have the potential to regulate co-translational protein folding mechanisms, and a growing number of proteins have been identified with folding mechanisms sensitive to codon usage. Typically, these proteins have complex folding pathways and kinetically stable native structures. Kinetically stable proteins may fold only once over their lifetime, and hence codon-mediated regulation of the “pioneer round” of protein folding can have a lasting impact. Supporting the important role of codon usage on folding, conserved patterns of codon usage appear in homologous gene families, hinting at selection. Despite these exciting developments, there is a paucity of experimental methods to quantify translation elongation rates and co-translational folding mechanisms in the cell, which challenges the development of a predictive understanding of how biology uses codons to regulate protein folding.

Keywords

synonymous codon substitution; co-translational protein folding; translation elongation; gene expression; functional protein production; rare codons

B. THE CONNECTION BETWEEN CODON USAGE AND PROTEIN FOLDING

Proper cell function is exquisitely sensitive to protein function, which is in turn exquisitely sensitive to the successful folding of proteins to their native, functional structures. Protein folding, including the avoidance of off-pathway (misfolded) conformations that can lead to aggregation, occurs during and after protein synthesis. Here we synthesize recent advances that have deepened our understanding of the impact of synonymous codon usage on protein folding mechanisms, including downstream effects on *fitness*.

For context, our current understanding of protein folding is dominated by results from an experimental approach pioneered by Christian Anfinsen more than 60 years ago, in which a purified, full-length protein is first unfolded in a chemical denaturant. Anfinsen found that with careful control of the experimental conditions, rapid dilution of ribonuclease

*Corresponding author, pclark1@nd.edu.

A (RNaseA) from denaturant can lead to spontaneous, reversible refolding to its native structure (48). This astounding discovery led to the current dominant paradigm that all information necessary to specify a protein native structure – amongst an astronomical number of other potential conformations – is encoded within the amino acid sequence. Given this paradigm, a necessary first question we must address is: How is it possible for synonymous codon substitutions, which do not alter the encoded amino acid sequence, to have any effect on protein folding?

The question above is especially timely right now. The development of increasingly sophisticated machine learning algorithms (9, 101), alongside an enormous and ever-growing repository of fully sequenced organisms, now enable accurate predictions of protein structure from amino acid sequence information alone (101). The accuracy of these predictions is a truly remarkable achievement – one that seemed far-fetched just a decade ago – and reinforces that most protein native structures represent an energy minimum specified solely by interactions between amino acids in the protein. However, these approaches are limited to native structure prediction: we are still unable to accurately predict the dominant mechanism used by a polypeptide on its journey from an unfolded state to its native structure (32). Subtle changes to proteins can lead to dramatic differences in folding mechanism; for example, the yeast and *E. coli* homologs of phosphoglycerate kinase have nearly identical structures and stabilities, yet their folding rates are 10⁵-fold different (121). Because we cannot yet accurately predict folding pathways, we cannot yet accurately predict how a co-translational folding mechanism will differ from the refolding of a full-length protein diluted from denaturant, although strides are being made in this direction (21, 57). However, what we do know is that the pathway taken for protein folding is often crucial for the success of the folding process.

The classic Anfinsen experiment has now been used to attempt to refold thousands of different proteins. Yet surprisingly, only a small fraction of proteins in the proteome have been found to refold reversibly, across a smooth, funnel-shaped energy landscape (Figure 1, top) (23). Most proteins, especially those that are large, multimeric, or have an otherwise complex structural topology (e.g., high contact order (95)), tend to misfold and aggregate rather than refold correctly when diluted from denaturant (113). Yet these proteins can fold robustly in the cell. This is a key observation, for two reasons. First, it means that there are multiple minima on the energy landscape for folding, including those that might lead to an irreversible “detour” to aggregation (Figure 1, middle). Indeed, Anfinsen himself recognized that the mechanism for RNaseA refolding upon dilution from denaturant was distinct from the mechanism used in the cell (5). The second important implication of the inability of most proteins to refold reversibly upon dilution from denaturant is that the cell must possess mechanisms to support proper folding in the challenging cellular environment. Anfinsen’s recognition of this reality spurred his discovery of one of the first molecular chaperones (45). However, addition of chaperones to the classic Anfinsen refolding experiment still leaves a significant fraction of proteins unable to refold (12, 114), indicating that chaperone availability alone is not sufficient for successful folding in the cellular environment.

One universal feature of the cellular environment for protein folding is that every protein in every cell can start to fold as it emerges from the exit tunnel of the ribosome

(Figure 2, blue). Co-translational folding at the rhythm of **elongation rate** is a distinctly different starting point for folding than refolding of full-length proteins upon dilution from denaturant. Returning to the question above: How is it possible for changes in **synonymous codon usage** to affect protein folding? The average rate of translation elongation is ~20 aa/sec in *E. coli* and ~4 aa/sec in mammals, but one key factor that modulates elongation rate is synonymous codon usage. Synonymous codon substitutions do not alter the amino acid sequence but can change the elongation rate of the ribosome by more than 10-fold (25, 124) (Sidebar 1). At an average rate of 20 aa/sec, synthesis of a 100 aa-long protein domain will take 5 s. Typical time constants for the folding of protein domains straddle this time but vary over many orders of magnitude, from msec to hr (110), and α -helices and β -sheets fold significantly faster (μ sec-msec and msec-sec, respectively) (27). These results have two important implications. First, N-terminal secondary and tertiary structures can start forming prior to the appearance of the protein C-terminus. Indeed, as detailed in Section C.2, many multimeric proteins form stable quaternary interactions while synthesis is still underway. Second, co-translational folding can be sensitive to changes to the rate of protein synthesis. For example, certain folding intermediates may not be populated during translation because the portions of a protein necessary for an interaction may not be synthesized quickly enough to prevent an alternative interaction from forming. Alternatively, slowing down elongation rate could provide an already-synthesized N-terminal portion of a nascent protein with more time to reach an equilibrium conformation, before introducing additional, potentially competing interactions with more C-terminal residues. Section C includes examples of how synonymous codon usage can alter protein folding mechanisms.

A second important question to address up front is: What is the significance of an altered mechanism for co-translational folding? In other words, if a protein can fold robustly to its native structure from essentially any starting conformation, and is sufficiently metastable that after folding co-translationally, it subsequently globally unfolds and refolds multiple times over its lifetime, does it matter if a different mechanism is used for the “pioneer round” of folding? For such proteins, the answer might indeed be: no, it does not matter. However, a significant fraction of the proteome consists of **kinetically stable** proteins, with native structures separated by a very large energy barrier from the unfolded ensemble (91, 120). For these proteins, protein folding likely occurs only once, and the details of co-translational protein folding mechanisms are likely crucial for navigating to the native structure (23, 34). More generally, co-translational folding can help avoid minima on the folding energy landscape that lead to misfolding and aggregation (Figure 1, bottom). Examples of these phenomena are explored in Sections C&D.

This review focuses on the impact of synonymous codon changes on **functional protein production**, specifically co-translational protein folding, multimeric protein assembly and membrane targeting and secretion. It builds on pivotal early results discussed in depth in prior reviews (52, 64, 89). Moreover, it is now clear that the coding sequences of homologous proteins tend to have conserved patterns of rare and common **synonymous codon usage** (30, 56), indicating that synonymous codon usage is under selection (see Section E). However, three decades of studying the impact of synonymous codons on co-translational folding has made our laboratory acutely aware of the numerous other cellular

mechanisms sensitive to synonymous codon substitutions, including transcription (127), pre-mRNA splicing (104), translation initiation (19, 71, 100), mRNA degradation (8), and translational fidelity (38, 44, 85) (Figure 2). Although not discussed in detail here, it is noteworthy that each of these mechanisms, while distinct from effects of elongation rate on folding, can also lead to changes in functional protein production, and hence complicates the detangling of the direct effects of synonymous codon substitutions on elongation rate and protein folding, versus other mechanisms (16, 29, 76). We end with a reflection on this and other current challenges and opportunities (Section F).

C. EFFECTS OF CODON USAGE ON THE FOLDING OF WATER-SOLUBLE PROTEINS

Stable contacts between amino acid residues in the N-terminal portion of the nascent polypeptide chain can begin to form once the polypeptide chain is outside of the ribosome, or even while still within the ribosome exit tunnel (2, 66, 78, 119) (Figure 3). The formation of co-translational folding intermediates reduces the nascent chain conformational ensemble, effectively selecting a subset of trajectories on its folding energy landscape (21, 34) (Figure 1, bottom). This reduction of conformational complexity could be especially beneficial to proteins with complex native structures, including those that are large, multimeric, or multi-domain, as these proteins tend to be more prone to misfolding and aggregation than small, single-domain proteins with low contact order. Organization of the emerging polypeptide chain into stable intermediate structures is further coordinated by molecular chaperones (12, 68), which can help reduce premature cleavage and degradation by proteases (23).

Although a growing number of studies have now shown that synonymous codon substitutions can affect co-translational folding pathways, it is important to acknowledge that the actual process of co-translational folding is currently very difficult to visualize and study directly, in part due to the rapid kinetics of both translation elongation and protein folding. For this reason, experimental studies of synonymous codon-induced changes to protein folding pathways are often detected indirectly, by assessing the impact on downstream parameters, including an altered native structure and/or solubility (Figure 3). By definition, for two proteins with the same sequence a change in the final structure must involve a change in the folding pathway (Figure 1, bottom), although not all perturbations to folding pathways will result in a change in the final structure. The assembly of multimeric proteins introduces additional complexity to co-translational folding. For this reason, the effects of synonymous codon substitutions on monomeric and multimeric proteins are presented below in two separate sub-sections.

C.1. Co-translational folding of monomeric proteins

The first direct evidence that synonymous codon substitutions can alter a protein folding pathway in a predictable way *in vivo* was found using the designed protein YKB (Yellow-blacK-Blue), which is based on split fluorescent proteins (103). YKB is a monomeric protein consisting of three half-domains, where the first and third half domains compete with each other to interact with the middle 'K' half-domain. Depending on which half-domain wins the competition, the result is either a yellow (YK) or blue (KB) fluorescent protein.

Crucial for this experiment, fluorescent proteins are kinetically stable: once either YK or KB forms, it does not equilibrate with the other structure. In other words, YKB only folds once over its lifetime (103). When YKB is diluted rapidly out of denaturant, it refolds to form equimolar amounts of the YK and KB structures. In contrast, when expressed in *E. coli*, yellow fluorescence increased and blue fluorescence decreased, indicating that, as expected, when folding can begin co-translationally, the N-terminal Y half-domain has a competitive advantage over B. Encoding the 5' end of the B half-domain with increasingly rare synonymous codons increasing the YK:KB fluorescence ratio as a function of codon rarity, presumably because the a slower appearance of B provided more time for Y to interact with K (103). The rare codon-induced decrease in local elongation rate effectively steered the YKB polypeptide chain towards one structure (YK) and away from another (KB).

While YKB demonstrated the potential for synonymous codon usage to affect folding pathways, as a designed protein it does not reveal whether or how changes to codon usage might affect co-translational folding in naturally occurring coding sequences. For example, YKB showed that slowing translation elongation with rare codons after the appearance of an entire protein domain can facilitate its folding prior to the appearance of an interfering C-terminal segment (103). Yet, broad analyses of synonymous codon conservation in bacteria, yeasts and across the tree of life indicate that while there is significant position-specific conservation of rare codon clusters in the coding sequences of homologous proteins, these conserved clusters are not enriched downstream of protein domain boundaries (30, 56, 93). Instead, these conserved clusters are found more often within protein domains, where they may coordinate co-translational folding at the sub-domain level.

As **codon optimization** (Sidebar 1) has become a standard tool for structural biology and drug development studies, a growing number of proteins have been identified that are sensitive to synonymous codon substitutions. For example, Alexaki *et al.* created an optimized version of human blood coagulation factor IX (FIX) using only very common synonymous codons, to increase elongation rate and **translational efficiency** (4). This optimized construct led to higher mRNA and protein levels. Yet while WT and optimized FIX had similar specific activity, differences were detected in FIX-specific antibody binding affinities and protease digestion patterns, suggesting subtle conformational differences between the two variants (4). These differences are most likely explained by a difference in elongation rate, which was supported by differences observed in ribosome profiling reads (4). Hence while codon optimization can increase gene expression, it can also affect the final structure of a protein due to an altered co-translational folding pathway.

Large-scale protein production strategies typically include expression in a heterologous host organism (88). However, proteins can fail to fold properly when expressed in a heterologous host, due to differences in genome **codon usage bias** (47, 58) (Sidebar 2). This phenomenon was studied in detail for translation of the bovine γ -B-crystallin gene on *E. coli* ribosomes, using either the original bovine coding sequence (unharmonized), or a **codon harmonized** variant where synonymous codons were selected based on similarities between bovine and *E. coli* codon usage frequencies (26). Proteinase K digestion revealed that, in contrast to the unharmonized version, the harmonized construct began to adopt stable structure at shorter

nascent chain lengths and folded to a more stable structure co-translationally (26). NMR was used to analyze the native structures of both variants, which confirmed that the synonymous substitutions affected the folding process (26).

Beyond designed proteins and heterologous gene expression, there is direct evidence that rare codons are necessary for proper folding and function in the native host organism. For example, the coding sequence for Cross Pathway Control Protein-1 (CPC-1) in *Neurospora crassa* (81) includes many rare U-rich codons (NNU), rather than the more commonly used (in *Neurospora*) NNC codons. In CPC-1, NNU codons are preferred over NNC for five amino acids. Mutating these NNU codons to synonymous NNC codons altered both the structure and function of CPC-1 (81). Specifically, the NNC mutant had increased mRNA and protein levels, indicating that wild type NNU codons suppress gene expression. Yet the introduction of NNC codons also altered both the stability and native structure of CPC-1, resulting in differences in both *in vivo* degradation rates and *in vitro* trypsin digestion rates, as well as adversely affecting CPC-1 activity and *Neurospora* growth rate (81). This study is a prime example of how nonoptimality (e.g., rare codons) can enhance protein function; introducing “optimal” codons is not always beneficial for protein function.

While kinetic stability can facilitate detecting changes in co-translational folding as changes in the native structure, it remains to be determined what minimum stability is necessary for codon-regulated elongation rate to significantly affect protein structure and cell function. Intriguing results in this direction come from studies of codon usage in FRQ, a circadian clock protein from *Neurospora* (125, 126). Most of FRQ is predicted to be intrinsically disordered and is encoded by low usage frequency codons. Optimization to more common codons led to an increase in FRQ abundance but abolished FRQ activity. Optimized FRQ was also more susceptible to protease digestion *in vitro*, had a shorter lifetime *in vivo*, and dysregulated circadian rhythms (125, 126). Hence changes to codon usage can lead to a persistent change in protein structure and function even for a protein that does not fold to a single stable structure.

Because co-translational folding intermediates are both short-lived and dynamic, they have proven challenging to study directly using current experimental methods. Computer simulations have helped fill gaps in this regard, generating predictions of co-translational folding intermediates. For example, all-atom Monte Carlo simulations have been used to study how vectorial appearance and codon usage can affect co-translational folding pathways (21). This study focused on three proteins (MarR, FabG, and CMK) that benefit from a reduction in elongation rate, which promotes the formation of co-translational folding intermediates and helps avoid non-native contacts. Hypotheses regarding the effects of synonymous codon usage on co-translational folding were also explored using a combination of Gō-based coarse-grained modelling and all-atom simulation studies of three model proteins: chloramphenicol acetyltransferase (CAT), D-alanine–D-alanine ligase B (DDLb) and dihydrofolate reductase (DHFR) (57). For each protein, elongation rate was increased or decreased by modeling variants encoded using either common or rare codons, respectively. As expected, for DHFR, which is known to refold robustly in the Anfinsen experiment (55), changing elongation rate had no effect on folding efficiency. In contrast, for CAT and DDLb, a heterologous mixture of long-lived “near native” misfolded states

were identified (57). For CAT and DDLB, these near native states are predicted to have similar catalytic efficiency, solubility, chaperone recruitment, and aggregation propensity. The computational results predicted a lasso-like entanglement leads to these kinetically trapped near native states, which was validated by limited proteolysis mass spectrometry (57). As with the two alternative native state scenario observed for YKB (103), described above, these long-lived states are a direct consequence of kinetic traps encountered – or avoided – during co-translational folding on the ribosome.

C.2. Assembly of multimeric proteins

Historically, most model proteins used to study folding have been monomeric. In contrast, most proteins in the proteome are multimeric – and rarely refold robustly to their native structures when diluted out of denaturant (23, 113). In contrast to monomeric proteins, multimeric proteins must not only fold correctly but also assemble with other subunits. Protein assembly is thus concentration dependent. Upon assembly, individual protein subunits can potentially undergo large conformational changes to form the native multimeric protein (83). Hence, depending on the native structure topology, assembly may occur before, during, and/or after folding of the individual polypeptide chains (102). Assembly that occurs before folding is complete can facilitate the formation of entangled native structures, in which the native structure of each subunit is critically dependent on interactions formed with neighboring subunits (3). Assembling with other subunits either too early or too late in folding can lead to trajectories that are off-pathway for formation of the native multimeric protein assembly, including mis-assembled soluble states (42) and aggregation (117). Just as co-translational folding can facilitate the folding of monomeric proteins to their native structure, co-translational assembly can coordinate proper folding for multimeric proteins (87). By coupling co-translational protein folding with assembly, cells can increase the folding yield and counteract the increased aggregation propensity of multimeric proteins (106).

All multimeric proteins can be classified into one of two groups: homomers, in which all subunits have the same amino acid sequence, and heteromers, which include subunits with different amino acid sequences. Both homomers and heteromers are found in all living organisms, although homomers dominate in prokaryotes while heteromers are more prevalent in eukaryotes (80). Broadly speaking, there are four possible assembly mechanisms for forming a multimeric protein (18, 87) (Figure 4). In eukaryotes, co-co cis is only compatible with homomeric protein assembly. In contrast, some prokaryotic mRNAs are **polycistronic**. Polycistronic mRNA opens the possibility of long-range co-co cis interactions formed between two subunits of a heteromeric complex encoded in a single mRNA (87), although it remains unclear how close the interaction interfaces would need to be encoded to one another in order to achieve a significant advantage in assembly efficiency versus co-post or post-post. In general, whether a multimeric protein will assemble co-translationally will depend on the affinity of partially synthesized polypeptide chains for one another and the position of the interacting segments along the nascent chain. Another crucial factor is the local concentration of the interacting segments within the nascent polypeptide chains, which is influenced by elongation rate and ribosome density (7). For co-co trans

assembly of heteromeric proteins, local mRNA concentrations can also affect the outcome (86, 87, 107).

Recently, hundreds of multimeric proteins (predominantly homodimers) were shown to assemble co-translationally using co-co mechanisms (either cis or trans) (18). These proteins were identified using disome selective profiling (DiSP), a new variation on ribosome profiling (54) that selectively analyzes the ribosome-protected mRNA fragment for ribosomes pairs that remain associated through strong intermolecular nascent chain interactions that persist through cell lysis, mRNA degradation and sucrose gradient centrifugation (18). The DiSP-identified proteins were enriched in coiled-coil and BAR (named after Bin, amphiphysin, and Rvs proteins) domains, located near the N-terminus (18). Although not tested by Bertolini et al., these co-translational interactions may be coordinated by synonymous codon usage. For example, rare codons that slow translation after synthesis interacting segments could increase the local density of nascent chains capable of interacting with one another on the same mRNA.

Co-localization of nascent chains to facilitate co-translational assembly has also been observed for heteromeric proteins. Groundbreaking early work from the lab of Tom Baldwin showed that bacterial luciferase must associate co-translationally to form the native $\alpha\beta$ heterodimer; without co-translational assembly, off-pathway $\beta\beta$ homodimers form (42). More recently, Günter Kramer, Bernd Bukau and coworkers tested the impact of gene proximity on successful luciferase assembly, by placing the α and β luciferase subunits either close together or far apart from one another on the bacterial chromosome (107). Each subunit was tagged with a fluorescent protein, and fluorescence resonance energy transfer (FRET) efficiency (as well as luciferase activity) was used to measure assembly of the native $\alpha\beta$ heterodimer. Native $\alpha\beta$ -luciferase assembly occurred more often when the genes encoding the α (LuxA) and β (LuxB) subunits were close together on the chromosome, presumably because co-locating these genes led to a higher local concentration of mRNA. Further, ribosome profiling showed that fully synthesized LuxA interacts with LuxB co-translationally only once the LuxB nascent chains are ~120 aa long, at which point the LuxB dimerization interface has emerged from the ribosome exit tunnel (107). These results emphasize that co-translational protein assembly is regulated both spatially and kinetically.

The astute reader will note that although both studies above suggest a potential role for regulation of multimeric protein assembly by synonymous codon usage, they do not include direct tests. Recently, however, it was shown that synonymous codon changes to the sequence encoding the homotrimeric *E. coli* antibiotic resistance protein chloramphenicol acetyltransferase (CAT) led to a subtly modified CAT native structure that is more susceptible to degradation by the *E. coli* protease ClpXP than wild type CAT (116). As with monomeric proteins, the formation of an altered final structure indicates a change in the folding pathway, presumably caused by the alteration of elongation rate via the synonymous codon substitutions, that enables access to a distinct energy minimum (Figure 1, bottom). *E. coli* expressing CAT from the alternative codon construct grew half as fast in the presence of chloramphenicol as *E. coli* expressing CAT from the wild type coding sequence, indicating the cellular chaperone network was unable to buffer the CAT folding pathway from these

changes (116). It remains to be determined if the effects of altering codon usage primarily affect folding of the CAT monomers, or assembly, or both.

D. EFFECTS OF CODON USAGE ON SECRETION AND TRANSMEMBRANE PROTEIN FOLDING

A significant fraction of every proteome (~30% in eukaryotes and bacteria; ~25% in archaea) is translocated through, integrated into or otherwise associated with a membrane (13, 49, 50, 82, 90). Proteins in the secretory pathway tend to have complex folding pathways. For example, most secreted proteins are integral membrane proteins with an α -helical fold (41, 69). Many of these proteins are translocated co-translationally and can begin to fold co-translationally (24, 49, 82, 90), and both co-translational folding and secretion can be affected by synonymous codon usage. Given their abundance in the cell and important roles in sensing and responding to the cellular environment, successful targeting and folding of secreted proteins have important implications for cellular fitness. In the following sub-sections, we review recent discoveries of how synonymous codons impact membrane targeting of proteins in the secretory pathway and co-translational folding of integral membrane proteins.

D1. Membrane targeting & intracellular trafficking

In eukaryotes, proteins in the secretory pathway are co-translationally targeted to the ER membrane by the signal recognition particle (SRP) (49, 60). Most secreted proteins contain an N-terminal signal sequence; once it emerges from the exit tunnel of the ribosome, the signal sequence binds to SRP (49). Although SRP has affinity for the ribosome itself, the signal sequence enhances SRP binding to the ribosome-nascent chain complex (RNC) (43). High-affinity binding of SRP to the RNC is followed by arrest of translation elongation and targeting of the arrested RNC-SRP complex to the Sec61 translocon in the ER membrane, via binding to the SRP receptor (17, 49, 60) (Figure 5). After binding the SRP receptor, elongation resumes, concomitant with translocation through Sec61 (49, 60). Water soluble proteins are translocated through the Sec61 pore into the ER lumen, while integral membrane proteins are transferred to the phospholipid bilayer by Sec61 and other insertion machinery (82, 109). SRP binding thus facilitates proper translocation, folding, and trafficking within the cell by coordinating translation elongation with targeting of the RNC to the translocon (73).

Synonymous codons have been shown to alter SRP-mediated targeting to the ER membrane for both secreted and membrane proteins, by modulating ribosome elongation rate during the critical early window of protein synthesis and SRP binding (31, 92) (Figure 5). Many transcripts encoding membrane and secreted proteins are enriched in non-optimal codons approximately forty codons after the SRP binding site (37, 92). This distance corresponds to the length of the ribosomal exit tunnel, suggesting that non-optimal codons in this position may lead to slower elongation as the SRP binding site emerges from the ribosome (92). Slow elongation at this position could provide more time for SRP to properly bind the RNC (Figure 5) (92). Alternatively, these slowly translated codons may be responsible for the elongation arrest that allows targeting of the RNC-SRP complex to the membrane

(31). Synonymous mutations to more optimal codons disrupt entry into the secretory pathway, presumably by increasing the elongation rate and thereby shortening the temporal window for SRP to engage the nascent chain or for RNC-SRP to arrive at the membrane (31, 92). However, many secreted proteins do not contain non-optimal codons at this position (31). It is currently unclear why some secreted proteins exhibit slow translation approximately 40 codons after the signal sequence while others do not. SRP engagement and trafficking of RNCs is therefore a complex process, but it appears synonymous codons, among other factors, can promote proper membrane targeting. Like water-soluble secreted proteins, integral membrane proteins are also enriched in slowly translated codons about 40 codons downstream of the first transmembrane α -helix, which serves as their signal sequence (92). The use of synonymous codons to modulate elongation rate is therefore a common mechanism for enhancing SRP binding across multiple classes of proteins within the secretory pathway.

Synonymous codons have also been shown to impact membrane targeting and cellular trafficking via SRP-independent pathways. Kesv is a viral protein that is structurally similar to eukaryotic potassium channels; it is typically targeted to the mitochondria when expressed in eukaryotic cells, but can be diverted to the ER by synonymous substitutions (39, 112). Cellular conditions and Kesv folding may play a role in its codon-dependent trafficking, though the precise mechanism has not been determined. Kesv localization to the mitochondria decreased and degradation increased when Kesv was translated and folded slowly at a reduced growth temperature compared to the optimal growth temperature, suggesting that Kesv folding influences its trafficking. Localization to the mitochondria was also affected by the stage of the cell cycle during targeting, suggesting that cell cycle-dependent factors are also involved in Kesv trafficking (39).

Like their eukaryotic counterparts, the coding sequences for *E. coli* secreted proteins are enriched in rare codons near the N-terminus, which may slow elongation rate and promote favorable interactions between the RNC and the membrane early during translation (35). Synonymous substitutions in the PelB signal sequence, used in bacteria to post-translationally target secreted proteins to the SecYEG translocon, have been shown to increase translocation efficiency (72). Synonymous mutations near the N-terminus and in the middle hydrophobic region of the signal sequence increased translocation relative to the wild type, though translocation was further enhanced by non-synonymous mutations that increased hydrophobicity (72). The effects of synonymous codons in other well-characterized post-translational signal sequences, such as PhoA, have not yet been investigated. Together, this work demonstrates that synonymous codons can modify the cellular trafficking of secreted proteins through several membrane-targeting pathways.

D.2. Integral membrane protein folding

In addition to N-terminal elongation arrest due to SRP binding, ribosomal pausing due to slowly translated codons has also been observed approximately 70 codons after subsequent transmembrane α -helices in membrane proteins that are co-translationally integrated into the membrane (59). Whether or how codon usage at these positions helps coordinate co-translational translocation and/or folding of membrane proteins remains unclear. However,

the spacing between the transmembrane α -helix and the suboptimal codons indicates the helix will have vacated the exit tunnel when translation is predicted to slow, so a ribosomal pause could facilitate engagement of the transmembrane domain with the bilayer, and/or membrane insertion machinery (59).

To date, most of our understanding of the effects of synonymous codon usage on membrane protein folding comes from studies of the cystic fibrosis transmembrane conductance regulator (CFTR), which belongs to the ABC transporter class of integral membrane proteins (Figure 6a) (14). Mutations and deletions in CFTR can cause cystic fibrosis (CF), in many cases due to CFTR misfolding (115). CFTR is a large protein comprised of 1,480 amino acids, and its synthesis is estimated to take seven to ten minutes (22), much longer than the time required to fold many protein domains, including the first nucleotide-binding domain (NBD1) of CFTR, where several disease-associated mutations reside (60). Like other α -helical membrane proteins, CFTR must fold and integrate into the lipid bilayer co-translationally (60), and its elongation rate appears to be non-uniform across the CFTR transcript (61, 96). Perhaps reflecting the CFTR native structural complexity and length of synthesis time, synonymous codon usage has been shown to affect CFTR translation, folding, and function. The broad suite of these effects has been reviewed elsewhere (14, 99, 123); here, we focus specifically on the impact of synonymous codons on CFTR co-translational folding.

The most common CF-causing mutation, $\Delta F508$, is an out-of-frame deletion in the codons encoding Ile507 and Phe508 in NBD1, which deletes Phe508 but also causes a synonymous change (ATC to ATT) to the Ile507 codon (Figure 6b) (15). The synergistic effect of these changes creates a folding defect that leads to CFTR degradation (14, 33, 79), although the native structure of $\Delta F508$ CFTR is largely unaffected by the mutation (79, 97). $\Delta F508$ slows elongation by modifying the mRNA secondary structure (15). Restoring the wild type Ile507 codon reverted the mRNA transcript to its wild type structure and rescued CFTR from degradation, even in the absence of Phe508 (15). Furthermore, the synonymous change to I507 may be responsible for the chloride channel gating defect in $\Delta F508$ CFTR, which contributes to its reduced activity at the plasma membrane (74). ATC and ATT codons occur with similar frequency in humans, suggesting that slow elongation of the ATT mutant may be due solely to its modified mRNA structure, rather than direct differences in elongation rate due to tRNA availability (15).

Other evidence supports a role for codon usage in CFTR co-translational folding by altering elongation rate. Various synonymous mutations have been shown to modulate CFTR elongation rate depending on the abundance of the corresponding tRNA (96). Furthermore, proper folding of the β -sheet core of NBD1 (Figure 6b) is dependent on the local elongation rate (22, 61). The elongation rate of the wild type sequence delays positioning of β -strand S6 until two subsequent β -strands emerge from the ribosome exit tunnel, allowing coordination of native contacts between the β -strands and an upstream α -subdomain. Synonymous mutations to common codons in this region caused misfolding of the β -strands, leading to aggregation (61). Because the NBD domain is conserved among ABC transporters, its sensitivity to elongation rate and codon usage for proper folding may be a common feature of proteins within this class.

Synonymous mutations can also act synergistically with non-synonymous mutations to modify CFTR folding. The synonymous mutation of Thr854 (Figure 6c) from ACT to ACG affects co-translational folding when it occurs in conjunction with other mutations (98). While this mutation alone is non-pathogenic, its effects on folding may carry important implications for disease phenotypes due to its high penetrance in the population (63). ACT to ACG is thought to slow translation elongation due to the relatively low concentration of the ACG iso-acceptor tRNA (63). This slow translation through the Thr854 ACG codon likely allows additional time for the alignment of domain interfaces, mitigating the disruption of domain interactions caused by nonsynonymous changes (98). Synonymous mutations can therefore impact multiple aspects of CFTR co-translational folding, including the proper formation of NBD1 structure and the establishment of interdomain interactions.

While most recent studies of codon usage in membrane proteins have focused on CFTR, synonymous mutations were also shown to disrupt folding of Pgp, another ABC transporter encoded by the *MDR1* gene (62). A Pgp variant containing one non-synonymous and two synonymous changes exhibited a modified conformation compared to the wild type, altering Pgp drug-pumping activity. While double mutants showed modest decreases in inhibition relative to wild type, the combination of all three mutations amplified this effect. The conformational change likely results from a decrease in the rate of translation elongation when I26 is mutated from ATC to the relatively rare ATT codon (62). An interesting open question is to what extent synonymous codon-induced folding changes are confined to proteins like CFTR and Pgp, versus occurring more broadly across other integral membrane protein types. Given the innate challenges of quantifying integral membrane protein folding, there likely remains much to discover.

E. IMPLICATIONS OF CODON USAGE EFFECTS FOR GENOME EVOLUTION

Historically, synonymous codon substitutions were viewed as genomic background noise (51). In the preceding sections, we highlighted recent studies showing how synonymous changes affect protein folding and fitness. The non-neutral impact of codon usage on co-translational folding, among other fitness effects, suggests that synonymous codons could be subject to selective pressure. Yet despite this mounting evidence, the synonymous mutation rate is still often considered to be equivalent to neutral drift (1, 38). The dN/dS ratio, calculated by normalizing the rate of nonsynonymous mutations to that of synonymous mutations, uses this assumption to detect deviations from neutral drift that would suggest selection on nonsynonymous mutations. It has been suggested that the use of the dN/dS ratio may not be an appropriate metric for determining selection, because synonymous mutations can be non-neutral (10, 28, 38).

Like all parts of the genome, protein-coding sequences accumulate mutations over time, and these mutations are subject to selective pressures to optimize fitness (75, 108). Synonymous mutations could therefore be subject to natural selection. Selection pressures to maintain productive co-translational folding may have resulted in conservation of codon usage patterns. Indeed, across the tree of life, position-specific analyses of codon usage within

the coding sequences of homologous proteins have revealed that rare codons often appear in conserved clusters (30) and a computational analysis has predicted that attenuating translation at these nascent chain lengths will facilitate formation of folding intermediates on pathway to the native structure (56). Similarly, an analysis of yeast genomes showed that both slowly and quickly translated codons are conserved and align with different types of secondary structures (93). Synonymous mutations can also cause misfolding and aggregation due to increased elongation rate, and the conservation of slowly translated codons suggests that there are selective pressures to avoid these outcomes (94). Together, these results suggest that elongation rate is conserved through codon usage to facilitate folding.

In addition to their individual effects on folding and fitness, synonymous mutations can also form epistatic interactions, where their effects on fitness are modified by other mutations that occur over the course of evolution. **Epistasis** can arise specifically from mutational effects on folding. For example, a survey of enzyme folding *in vivo* demonstrated that mutations exhibiting epistatic interactions could have either beneficial or deleterious effects on fitness depending on the codon (40). Epistatic interactions have also been observed between synonymous and non-synonymous mutations (98). A synonymous substitution in CFTR was shown to exhibit positive epistasis by rescuing a misfolded nonsynonymous variant (98). Epistatic interactions involving synonymous codons can impact protein folding and fitness and may therefore help shape evolutionary pathways. Further investigations are needed to determine the prevalence of these interactions in other classes of proteins.

F. FUTURE OPPORTUNITIES

Although much progress has been made, there remain several significant barriers to the development of a clear picture of what specific changes occur to a co-translational protein folding pathway when elongation rate changes. These barriers represent opportunities to advance the field in a significant way. First, we lack an experimental method capable of selectively reporting on subtle structural changes in nascent chain folding intermediates on the rapid timescale of protein synthesis. To be clear, such a method would most likely need to work both (a) as a single-molecule experiment, to overcome the inherent stochasticity of bulk translation reactions, and (b) in cells, to preserve elongation rates that are relevant to folding rates.

A second challenge is correlating changes in folding to changes in elongation rate without an accurate model of absolute elongation rates *in vivo*. While we know that some codons tend to slow down elongation rate due to tRNA scarcity, or wobble base mismatch, larger-scale effects like dicodons and the impact of mRNA structure on elongation rate are still challenging to predict (25). This is especially true in cells, where polyribosomes are expected to have a significant impact on mRNA structure.

A third challenge is that synonymous codon substitutions can affect many aspects of functional protein production upstream of co-translational folding. Detangling effects specific to folding from those related to other aspects of protein production can be extremely challenging, especially because such effects can be subtle and/or cumulative. Indeed, the

recent claim of strong, broad negative effects of synonymous mutations on fitness in yeast (105) were immediately viewed with skepticism (and ultimately attributed to issues with experimental design (70)) in part because our cumulative experience with synonymous and non-synonymous mutations continues to support that the effects of synonymous mutations will mostly be more subtle than non-synonymous mutations. Nevertheless, the growing list of disease-associated synonymous mutations (51) indicates that these effects, even when subtle, can have a significant impact on fitness.

ACKNOWLEDGEMENTS

This work was supported by a grant from the US National Institutes of Health (DP1 GM146256). M.J.M. is supported by a Graduate Research Fellowship from the National Science Foundation.

TERMS AND DEFINITIONS

Codon harmonization

adjusting codon usage to match usage frequencies (or other codon characteristics) from an origin organism in a heterologous host

Codon optimization

increasing gene expression by using synonymous codons enriched in highly expressed genes

Codon usage bias

unequal usage of synonymous codons, at the intra-gene, gene and/or genome levels

Elongation rate

average speed of synthesis for one peptide bond by the ribosome

Epistasis

phenomenon where the effect of a mutation on fitness is modified by other mutations

Fitness

ability of an organism to develop and reproduce

Functional protein production

the combination of gene expression (transcription and translation) and successful protein folding

Kinetically stable

separation of native and denatured states by such a large barrier that a protein effectively never unfolds, regardless of thermodynamic stability

Polycistronic

a type of prokaryotic mRNA that includes two or more coding sequences

Translational efficiency

ratio of the number of protein molecules synthesized per mRNA molecule

REFERENCES

1. Agashe D, Sane M, Singhal S. 2023. Revisiting the role of genetic variation in adaptation. *Am Nat*
2. Agirrezabala X, Samatova E, Macher M, Liutkute M, Maiti M, et al. 2022. A switch from α -helical to β -strand conformation during co-translational protein folding. *EMBO J.* 41(4):1–13
3. Ahnert SE, Marsh JA, Hernández H, Robinson CV, Teichmann SA. 2015. Principles of assembly reveal a periodic table of protein complexes. *Science.* 350(6266):aaa22451–510
4. Alexaki A, Hettiarachchi GK, Athey JC, Katneni UK, Simhadri V, et al. 2019. Effects of codon optimization on coagulation factor IX translation and structure: Implications for protein and gene therapies. *Sci Rep.* 9(1):1–15 [PubMed: 30626917]
5. Anfinsen CB. 1973. Principles that govern the folding of protein chains. *Science.* 181(4096):223–30 [PubMed: 4124164]
6. Angov E, Hillier CJ, Kincaid RL, Lyon JA. 2008. Heterologous protein expression is enhanced by harmonizing the codon usage frequencies of the target gene with those of the expression host. *PLoS One.* 3(5):1–10
7. Badonyi M, Marsh JA. 2023. Hallmarks and evolutionary drivers of cotranslational protein complex assembly. *FEBS Journal*, pp. 1–11
8. Bae H, Collier J. 2022. Codon optimality-mediated mRNA degradation: Linking translational elongation to mRNA stability. *Mol Cell.* 82(8):1467–76 [PubMed: 35452615]
9. Baek M, DiMaio F, Anishchenko I, Dauparas J, Ovchinnikov S, et al. 2021. Accurate prediction of protein structures and interactions using a three-track neural network. *Science.* 373(6557):871–76 [PubMed: 34282049]
10. Bailey SF, Morales LAA, Kassen R. 2021. Effects of Synonymous Mutations beyond Codon Bias: The Evidence for Adaptive Synonymous Substitutions from Microbial Evolution Experiments. *Genome Biol Evol.* 13(9):1–18
11. Baker D, Agard DA. 1994. Kinetics versus Thermodynamics in Protein Folding. *Biochemistry.* 33(24):7505–9 [PubMed: 8011615]
12. Balchin D, Hayer-Hartl M, Hartl FU. 2016. In vivo aspects of protein folding and quality control. *Science.* 353(6294):aac43541–412
13. Bardy SL, Eichler J, Jarrell KF. 2003. Archaeal signal peptides--a comparative survey at the genome level. *Protein Sci.* 12(9):1833–43 [PubMed: 12930983]
14. Bartoszewski R, Króliczewski J, Piotrowski A, Jasiecka AJ, Bartoszewska S, et al. 2016. Codon bias and the folding dynamics of the cystic fibrosis transmembrane conductance regulator. *Cell Mol Biol Lett.* 21(1):1–13 [PubMed: 28536604]
15. Bartoszewski RA, Jablonsky M, Bartoszewska S, Stevenson L, Dai Q, et al. 2010. A synonymous single nucleotide polymorphism in $\Delta F508$ CFTR alters the secondary structure of the mRNA and the expression of the mutant protein. *J Biol Chem.* 285(37):28741–48 [PubMed: 20628052]
16. Bergman S, Tuller T. 2020. Widespread non-modular overlapping codes in the coding regions. *Phys Biol.* 17(3):1–15
17. Berndt U, Oellerer S, Zhang Y, Johnson AE, Rospert S. 2009. A signal-anchor sequence stimulates signal recognition particle binding to ribosomes from inside the exit tunnel. *PNAS.* 106(5):1398–1403 [PubMed: 19164516]
18. Bertolini M, Fenzl K, Kats I, Wruck F, Tippmann F, et al. 2021. Interactions between nascent proteins translated by adjacent ribosomes drive homomer assembly. *Science.* 371(6524):57–64 [PubMed: 33384371]
19. Bhattacharyya S, Jacobs WM, Adkar BV, Yan J, Zhang W, Shakhnovich EI. 2018. Accessibility of the Shine-Dalgarno Sequence Dictates N-Terminal Codon Bias in *E. coli*. *Mol Cell.* 70(5):894–905.e5 [PubMed: 29883608]
20. Bhattacharyya S, Jacobs WM, Adkar BV, Yan J, Zhang W, Shakhnovich EI. 2018. Accessibility of the Shine-Dalgarno Sequence Dictates N-Terminal Codon Bias in *E. coli*. *Mol Cell.* 70(5):894–905 [PubMed: 29883608]
21. Bitran A, Jacobs WM, Zhai X, Shakhnovich E. 2020. Cotranslational folding allows misfolding-prone proteins to circumvent deep kinetic traps. *PNAS.* 117(3):1485–95 [PubMed: 31911473]

22. Bose SJ, Krainer G, Ng DRS, Schenkel M, Shishido H, et al. 2020. Towards next generation therapies for cystic fibrosis: Folding, function and pharmacology of CFTR. *J Cyst Fibros.* 19:S25–32 [PubMed: 31902693]
23. Braselmann E, Chaney JL, Clark PL. 2013. Folding the proteome. *Trends Biochem Sci.* 38(7):337–44 [PubMed: 23764454]
24. Brodsky JL, Skach WR. 2011. Protein folding and quality control in the endoplasmic reticulum: Recent lessons from yeast and mammalian cell systems. *Curr Opin Cell Biol.* 23(4):464–75 [PubMed: 21664808]
25. Brule CE, Grayhack EJ. 2017. Synonymous Codons: Choose Wisely for Expression. *Trends Genet.* 33(4):283–97 [PubMed: 28292534]
26. Buhr F, Jha S, Thommen M, Mittelstaet J, Kutz F, et al. 2016. Synonymous Codons Direct Cotranslational Folding toward Different Protein Conformations. *Mol Cell.* 61(3):341–51 [PubMed: 26849192] --- codon harmonization emphasizes elongation rate is important in reaching final structure
27. Campos LA, Sadqi M, Muñoz V. 2020. Lessons about Protein Folding and Binding from Archetypal Folds. *Acc Chem Res.* 53(10):2180–88 [PubMed: 32914959]
28. Chandra S, Gupta K, Khare S, Kohli P, Asok A, et al. 2022. The High Mutational Sensitivity of ccdA Antitoxin Is Linked to Codon Optimality. *Mol Biol Evol.* 39(10):1–21
29. Chaney JL, Clark PL. 2015. Roles for Synonymous Codon Usage in Protein Biogenesis. *Annu Rev Biophys.* 44:143–66 [PubMed: 25747594]
30. Chaney JL, Steele A, Carmichael R, Rodriguez A, Specht AT, et al. 2017. Widespread position-specific conservation of synonymous rare codons within coding sequences. *PLoS Comput Biol.* 13(5):1–19 --- identified position-specific conservation of codon usage across the tree of life, but not at domain boundaries
31. Chartron JW, Hunt KCL, Frydman J. 2016. Cotranslational signal-independent SRP preloading during membrane targeting. *Nature.* 536(7615):224–28 [PubMed: 27487213]
32. Chen SJ, Hassan M, Jernigan RL, Jia K, Kihara D, et al. 2023. Protein folds vs. protein folding: Differing questions, different challenges. *PNAS.* 120(1):1–4
33. Cheng SH, Gregory RJ, Marshall J, Paul S, Souza DW, et al. 1990. Defective Intracellular Transport and Processing of CFTR Is the Molecular Basis of Most Cystic Fibrosis. *Cell.* 63:827–34 [PubMed: 1699669]
34. Clark PL. 2004. Protein folding in the cell: reshaping the folding funnel. *Trends Biochem Sci.* 29(10):527–34 [PubMed: 15450607]
35. Clarke TF, Clark PL. 2010. Increased incidence of rare codon clusters at 5' and 3' gene termini: implications for function. *BMC Genomics.* 11(118):1–10 [PubMed: 20044946]
36. Costantini M, Musto H. 2017. The Isochores as a Fundamental Level of Genome Structure and Organization: A General Overview. *J Mol Evol.* 84(2–3):93–103 [PubMed: 28243687]
37. Dessen P, Képè F. 2000. The PAUSE software for analysis of translational control over protein targeting: application to *E. nidulans* membrane proteins. *Gene.* 244:89–96 [PubMed: 10689191]
38. Drummond DA, Wilke CO. 2008. Mistranslation-Induced Protein Misfolding as a Dominant Constraint on Coding-Sequence Evolution. *Cell.* 134(2):341–52 [PubMed: 18662548]
39. Engel AJ, Kithil M, Langhans M, Rauh O, Cartolano M, et al. 2021. Codon bias can determine sorting of a potassium channel protein. *Cells.* 10(5):1–17
40. Faber MS, Wrenbeck EE, Azouz LR, Steiner PJ, Whitehead TA. 2019. Impact of in Vivo Protein Folding Probability on Local Fitness Landscapes. *Mol Biol Evol.* 36(12):2764–77 [PubMed: 31400199]
41. Fagerberg L, Jonasson K, Von Heijne G, Uhlén M, Berglund L. 2010. Prediction of the human membrane proteome. *Proteomics.* 10(6):1141–49 [PubMed: 20175080]
42. Fedorov AN, Baldwin TO. 1995. Contribution of cotranslational folding to the rate of formation of native protein structure. *PNAS.* 92(4):1227–31 [PubMed: 7862665]
43. Flanagan JJ, Chen JC, Miao Y, Shao Y, Lin J, et al. 2003. Signal recognition particle binds to ribosome-bound signal sequences with fluorescence-detected subnanomolar affinity that does not diminish as the nascent chain lengthens. *J Biol Chem.* 278(20):18628–37 [PubMed: 12621052]

44. Garofalo R, Wohlgemuth I, Pearson M, Lenz C, Urlaub H, Rodnina MV. 2019. Broad range of missense error frequencies in cellular proteins. *Nucleic Acids Res.* 47(6):2932–45 [PubMed: 30649420]
45. Goldberger RF, Epstein CJ, Anfinsen CB. 1963. Acceleration of reactivation of reduced bovine pancreatic ribonuclease by a microsomal system from rat liver. *J Biol Chem.* 238(2):628–35 [PubMed: 13948694]
46. Grantham R, Gautier C, Gouy M, Jacobzone M, Mercier R. 1981. Codon catalog usage is a genome strategy modulated for gene expressivity. *Nucleic Acids Res.* 9(1):43–74
47. Gustafsson C, Govindarajan S, Minshull J. 2004. Codon bias and heterologous protein expression. *Trends Biotechnol.* 22(7):346–53 [PubMed: 15245907]
48. Haber E, Anfinsen CB. 1961. Regeneration of Enzyme Activity by Air Oxidation of Reduced Subtilisin-Modified Ribonuclease. *J Biol Chem.* 236(2):422–24 [PubMed: 13710229]
49. Hsieh HH, Shan SO. 2022. Fidelity of cotranslational protein targeting to the endoplasmic reticulum. *Int J Mol Sci.* 23(1):1–15
50. Hui X, Chen Z, Zhang J, Lu M, Cai X, et al. 2021. Computational prediction of secreted proteins in gram-negative bacteria. *Comput Struct Biotechnol J.* 19:1806–28 [PubMed: 33897982]
51. Hunt RC, Kimchi-Sarfaty C. 2022. When Silence Disrupts. *N Engl J Med.* 387(8):753–56 [PubMed: 36001717]
52. Hunt RC, Simhadri VL, Iandoli M, Sauna ZE, Kimchi-Sarfaty C. 2014. Exposing synonymous mutations. *Trends in Genetics.* 30(7):308–21 [PubMed: 24954581]
53. Ikemura T. 1981. Correlation between the Abundance of Escherichia coli Transfer RNAs and the Occurrence of the Respective Codons in its Protein Genes: A Proposal for a Synonymous Codon Choice that is Optimal for the E. coli Translational System. *J. Mol. Biol.* 151:389–409
54. Ingolia NT, Ghaemmamghami S, Newman JRS, Weissman JS. 2009. Genome-wide analysis in vivo of translation with nucleotide resolution using ribosome profiling. *Science.* 324(5924):218–23 [PubMed: 19213877]
55. Ionescu RM, Smith VF, O'Neill JC, Matthews CR. 2000. Multistate equilibrium unfolding of Escherichia coli dihydrofolate reductase: thermodynamic and spectroscopic description of the native, intermediate, and unfolded ensembles. *Biochemistry.* 39(31):9540–50 [PubMed: 10924151]
56. Jacobs WM, Shakhnovich EI. 2017. Evidence of evolutionary selection for cotranslational folding. *PNAS.* 114(43):11434–39 [PubMed: 29073068] --- pairs identification of conserved rare codon clusters with co-translational folding predictions
57. Jiang Y, Neti SS, Sitarik I, Pradhan P, To P, et al. 2023. How synonymous mutations alter enzyme structure and function over long timescales. *Nat Chem.* 15(3):308–18 [PubMed: 36471044]
58. Kane JF. 1995. Effects of rare codon clusters on high-level expression of heterologous proteins in Escherichia coli. *Curr Opin Biotechnol.* 6(5):494–500 [PubMed: 7579660]
59. Kepes F. 1996. The “+70 Pause”: Hypothesis of a Translational Control of Membrane Protein Assembly. *J Mol Biol.* 1996:77–86
60. Kim SJ, Skach WR. 2012. Mechanisms of CFTR Folding at the Endoplasmic Reticulum. *Front Pharmacol.* 3:1–11 [PubMed: 22291651]
61. Kim SJ, Yoon JS, Shishido H, Yang Z, Rooney LA, et al. 2015. Translational tuning optimizes nascent protein folding in cells. *Science.* 348(6233):444–48 [PubMed: 25908822] --- codons modulate translation elongation rate to support folding of an integral membrane protein
62. Kimchi-Sarfaty C, Oh JM, Kim I-W, Sauna ZE, Calcagno AM, et al. 2007. A “silent” polymorphism in the MDR1 gene changes substrate specificity. *Science.* 315(5811):525–28 [PubMed: 17185560]
63. Kirchner S, Cai Z, Rauscher R, Kastelic N, Anding M, et al. 2017. Alteration of protein function by a silent polymorphism linked to tRNA abundance. *PLoS Biol.* 15(5):1–29
64. Komar AA. 2009. A pause for thought along the co-translational folding pathway. *Trends Biochem Sci.* 34(1):16–24 [PubMed: 18996013]
65. Komar AA. 2021. A Code Within a Code: How Codons Fine-Tune Protein Folding in the Cell. *Biochemistry (Mosc).* 86(8):976–91 [PubMed: 34488574]

66. Kosolapov A, Deutsch C. 2009. Tertiary interactions within the ribosomal exit tunnel. *Nat Struct Mol Biol.* 16(4):405–11 [PubMed: 19270700]
67. Koutmou KS, Radhakrishnan A, Green R. 2015. Synthesis at the Speed of Codons. *Trends Biochem Sci.* 40(12):717–18 [PubMed: 26526516]
68. Kramer G, Shiber A, Bukau B. 2019. Mechanisms of cotranslational maturation of newly synthesized proteins. *Annu Rev Biochem.* 88:337–64 [PubMed: 30508494]
69. Krogh A, Larsson B, Von Heijne G, Sonnhammer ELL. 2001. Predicting transmembrane protein topology with a hidden Markov model: Application to complete genomes. *J Mol Biol.* 305(3):567–80 [PubMed: 11152613]
70. Kruglyak L, Beyer A, Bloom JS, Grossbach J, Lieberman TD, et al. 2023. Insufficient evidence for non-neutrality of synonymous mutations. *Nature.* 616(7957):E8–9 [PubMed: 37076734]
71. Kudla G, Murray AW, Tollervey D, Plotkin JB. 2009. Coding-sequence determinants of gene expression in *Escherichia coli*. *Science.* 324(5924):255–58 [PubMed: 19359587]
72. Kulmala A, Lappalainen M, Lamminmäki U, Huovinen T. 2022. Synonymous Codons and Hydrophobicity Optimization of Post-translational Signal Peptide PelB Increase Phage Display Efficiency of DARPins. *ACS Synth Biol.* 11(10):3174–81 [PubMed: 36178799]
73. Lakkaraju AKK, Mary C, Scherrer A, Johnson AE, Strub K. 2008. SRP Keeps Polypeptides Translocation-Competent by Slowing Translation to Match Limiting ER-Targeting Sites. *Cell.* 133(3):440–51 [PubMed: 18455985]
74. Lazrak A, Fu L, Bali V, Bartoszewski R, Rab A, et al. 2013. The silent codon change I507-ATC->ATT contributes to the severity of the Δ F508 CFTR channel dysfunction. *FASEB J.* 27(11):4630–45 [PubMed: 23907436]
75. Liberles DA, Teichmann SA, Bahar I, Bastolla U, Bloom J, et al. 2012. The interface of protein structure, protein biophysics, and molecular evolution. *Protein Sci.* 21(6):769–85 [PubMed: 22528593]
76. Liu Y, Yang Q, Zhao F. 2021. Synonymous but Not Silent: The Codon Usage Code for Gene Expression and Protein Folding. *Annu Rev Biochem.* 90:375–401 [PubMed: 33441035]
77. Livingston NM, Kwon J, Valera O, Saba JA, Sinha NK, et al. 2023. Bursting translation on single mRNAs in live cells. *Mol Cell.* 83:1–14 [PubMed: 36608666]
78. Lu J, Deutsch C. 2005. Folding zones inside the ribosomal exit tunnel. *Nat Struct Mol Biol.* 12(12):1123–29 [PubMed: 16299515]
79. Lukacs GL, Verkman AS. 2012. CFTR: Folding, misfolding and correcting the Δ F508 conformational defect. *Trends Mol Med.* 18(2):81–91 [PubMed: 22138491]
80. Lynch M. 2012. The evolution of multimeric protein assemblages. *Mol Biol Evol.* 29(5):1353–66 [PubMed: 22144639]
81. Lyu X, Liu Y. 2020. Nonoptimal codon usage is critical for protein structure and function of the master general amino acid control regulator CPC-1. *mBio.* 11(5):1–11 --- synonymous codon substitutions affect folding and fitness in *Neurospora*
82. Marinko JT, Huang H, Penn WD, Capra JA, Schleich JP, Sanders CR. 2019. Folding and Misfolding of Human Membrane Proteins in Health and Disease: From Single Molecules to Cellular Proteostasis. *Chem Rev.* 119(9):5537–5606 [PubMed: 30608666]
83. Marsh JA, Teichmann SA. 2011. Relative solvent accessible surface area predicts protein conformational changes upon binding. *Structure.* 19(6):859–67 [PubMed: 21645856]
84. Mignon C, Mariano N, Stadthagen G, Lugari A, Lagoutte P, et al. 2018. Codon harmonization – going beyond the speed limit for protein expression. *FEBS Lett.* 592(9):1554–64 [PubMed: 29624661]
85. Mordret E, Dahan O, Asraf O, Rak R, Yehonadav A, et al. 2019. Systematic Detection of Amino Acid Substitutions in Proteomes Reveals Mechanistic Basis of Ribosome Errors and Selection for Translation Fidelity. *Mol Cell.* 75(3):427–441.e5 [PubMed: 31353208]
86. Natan E, Endoh T, Haim-Vilmovsky L, Flock T, Chalancon G, et al. 2018. Cotranslational protein assembly imposes evolutionary constraints on homomeric proteins. *Nat Struct Mol Biol.* 25(3):279–88 [PubMed: 29434345]
87. Natan E, Wells JN, Teichmann SA, Marsh JA. 2017. Regulation, evolution and consequences of cotranslational protein complex assembly. *Curr Opin Struct Biol.* 42:90–97 [PubMed: 27969102]

88. Nieuwkoop T, Finger-Bou M, van der Oost J, Claassens NJ. 2020. The Ongoing Quest to Crack the Genetic Code for Protein Production. *Mol Cell*. 80(2):193–209 [PubMed: 33010203]
89. O'Brien EP, Ciryam P, Vendruscolo M, Dobson CM. 2014. Understanding the influence of codon translation rates on cotranslational protein folding. *Acc Chem Res*. 47(5):1536–44 [PubMed: 24784899]
90. Oswald J, Njenga R, Natriashvili A, Sarmah P, Koch HG. 2021. The Dynamic SecYEG Translocon. *Front Mol Biosci*. 8:1–30
91. Park C, Zhou S, Gilmore J, Marqusee S. 2007. Energetics-based Protein Profiling on a Proteomic Scale: Identification of Proteins Resistant to Proteolysis. *J Mol Biol*. 368:1426–37 [PubMed: 17400245]
92. Pechmann S, Chartron JW, Frydman J. 2014. Local slowdown of translation by nonoptimal codons promotes nascent-chain recognition by SRP in vivo. *Nat Struct Mol Biol*. 21(12):1100–1105 [PubMed: 25420103]
93. Pechmann S, Frydman J. 2013. Evolutionary conservation of codon optimality reveals hidden signatures of cotranslational folding. *Nat Struct Mol Biol*. 20(2):237–43 [PubMed: 23262490]
94. Perach M, Zafir Z, Tuller T, Lewinson O. 2021. Identification of conserved slow codons that are important for protein expression and function. *RNA Biol*. 18(12):2296–2307 [PubMed: 33691590]
95. Plaxco KW, Simons KT, Baker D. 1998. Contact Order, Transition State Placement and the Refolding Rates of Single Domain Proteins. *J Mol Biol*. 277:985–94 [PubMed: 9545386]
96. Polte C, Wedemeyer D, Oliver KE, Wagner J, Bijvelds MJC, et al. 2019. Assessing cell-specific effects of genetic variations using tRNA microarrays. *BMC Genomics*. 20:1–12 [PubMed: 30606130]
97. Qu B-H, Thomas PJ. 1996. Alteration of the Cystic Fibrosis Transmembrane Conductance Regulator Folding Pathway. *J Biol Chem*. 271(13):7261–64 [PubMed: 8631737]
98. Rauscher R, Bampi GB, Guevara-Ferrer M, Santos LA, Joshi D, et al. 2021. Positive epistasis between disease-causing missense mutations and silent polymorphism with effect on mRNA translation velocity. *PNAS*. 118(4):1–8 --- synonymous mutations can act synergistically with non-synonymous mutations to affect protein folding
99. Rauscher R, Ignatova Z. 2018. Timing during translation matters: Synonymous mutations in human pathologies influence protein folding and function. *Biochem Soc Trans*. 46(4):937–44 [PubMed: 30065107]
100. Rojano-Nisimura AM, Haning K, Janovsky J, Vasquez KA, Thompson JP, Contreras LM. 2020. Codon Selection Affects Recruitment of Ribosome-Associating Factors during Translation. *ACS Synth Biol*. 9(2):329–42 [PubMed: 31769967]
101. Ronneberger O, Tunyasuvunakool K, Bates R, Židek A, Ballard AJ, et al. 2021. Highly accurate protein structure prediction with AlphaFold. *Nature*. 596:583–89 [PubMed: 34265844]
102. Rumfeldt JAO, Vassall KA, Meiering EM. 2008. Conformational stability and folding mechanisms of dimeric proteins. *Prog Biophys Mol Biol*. 98:61–68 [PubMed: 18602415]
103. Sander IM, Chaney JL, Clark PL. 2014. Expanding Anfinsen's principle: contributions of synonymous codon selection to rational protein design. *J Am Chem Soc*. 136(3):858–61 [PubMed: 24392935] --- first direct evidence that synonymous substitutions can predictably affect a protein folding mechanism in vivo
104. Sarkar A, Panati K, Narala VR. 2022. Code inside the codon: The role of synonymous mutations in regulating splicing machinery and its impact on disease. *Mutat Res Rev Mutat Res*. 790:1–17
105. Shen X, Song S, Li C, Zhang J. 2022. Synonymous mutations in representative yeast genes are mostly strongly non-neutral. *Nature*. 606(7915):725–31 [PubMed: 35676473]
106. Shiber A, Döring K, Friedrich U, Klann K, Merker D, et al. 2018. Cotranslational assembly of protein complexes in eukaryotes revealed by ribosome profiling. *Nature*. 561(7722):268–72 [PubMed: 30158700]
107. Shieh Y-W, Minguez P, Bork P, Auburger JJ, Guilbride DL, et al. 2015. Operon structure and cotranslational subunit association direct protein assembly in bacteria. *Science*. 350(6261):678–80 [PubMed: 26405228]
108. Sikosek T, Chan HS. 2014. Biophysics of protein evolution and evolutionary protein biophysics. *J R Soc Interface*. 11(100):1–35

109. Smalinskaitė L, Hegde RS. 2023. The Biogenesis of Multipass Membrane Proteins. *Cold Spring Harb Perspect Biol.* 15(4):1–21
110. Sosnick TR, Barrick D. 2011. The folding of single domain proteins--have we reached a consensus? *Curr Opin Struct Biol.* 21(1):12–24 [PubMed: 21144739]
111. Spencer PS, Siller E, Anderson JF, Barral JM. 2012. Silent substitutions predictably alter translation elongation rates and protein folding efficiencies. *J Mol Biol.* 422(3):328–35 [PubMed: 22705285]
112. Thiel G, Baumeister D, Schroeder I, Kast SM, Van Etten JL, Moroni A. 2011. Minimal art: Or why small viral K⁺ channels are good tools for understanding basic structure and function relations. *Biochim Biophys Acta Biomembr.* 1808(2):580–88
113. To P, Whitehead B, Tarbox HE, Fried SD. 2021. Nonrefoldability is Pervasive across the E. coli Proteome. *J Am Chem Soc.* 143(30):11435–48 [PubMed: 34308638]
114. To P, Xia Y, Lee SO, Devlin T, Fleming KG, Fried SD. 2022. A proteome-wide map of chaperone-assisted protein refolding in a cytosol-like milieu. *PNAS.* 119(48):1–12
115. Veit G, Avramescu RG, Chiang AN, Houck SA, Cai Z, et al. 2016. From CFTR biology toward combinatorial pharmacotherapy: Expanded classification of cystic fibrosis mutations. *Mol Biol Cell.* 27(3):424–33 [PubMed: 26823392]
116. Walsh IM, Bowman MA, Soto Santarriaga IF, Rodriguez A, Clark PL. 2020. Synonymous codon substitutions perturb cotranslational protein folding in vivo and impair cell fitness. *PNAS.* 117(7):3528–34 [PubMed: 32015130] --- synonymous codon substitutions alters folding of a homotrimeric protein, decreasing fitness
117. Wright CF, Teichmann SA, Clarke J, Dobson CM. 2005. The importance of sequence diversity in the aggregation and evolution of proteins. *Nature.* 438(7069):878–81 [PubMed: 16341018]
118. Wright G, Rodriguez A, Li J, Milenkovic T, Emrich SJ, Clark PL. 2022. CHARMING: Harmonizing synonymous codon usage to replicate a desired codon usage pattern. *Protein Sci.* 31(1):221–31 [PubMed: 34738275]
119. Wruck F, Tian P, Kudva R, Best RB, von Heijne G, et al. 2021. The ribosome modulates folding inside the ribosomal exit tunnel. *Commun Biol.* 4(1):1–8 [PubMed: 33398033]
120. Xia K, Manning M, Hesham H, Lin Q, Bystroff C, Coló W. 2007. Identifying the subproteome of kinetically stable proteins via diagonal 2D SDS/PAGE. *PNAS.* 104(44):17329–34 [PubMed: 17956990]
121. Young TA, Skordalakes E, Marqusee S. 2007. Comparison of Proteolytic Susceptibility in Phosphoglycerate Kinases from Yeast and E. coli: Modulation of Conformational Ensembles Without Altering Structure or Stability. *J Mol Biol.* 368(5):1438–47 [PubMed: 17397866]
122. Yu CH, Dang Y, Zhou Z, Wu C, Zhao F, et al. 2015. Codon Usage Influences the Local Rate of Translation Elongation to Regulate Co-translational Protein Folding. *Mol Cell.* 59(5):744–54 [PubMed: 26321254]
123. Zhang Y, Bebok Z. 2022. An Examination of Mechanisms by which Synonymous Mutations may Alter Protein Levels, Structure and Functions. In *Single Nucleotide Polymorphisms: Human Variation and a Coming Revolution in Biology and Medicine*, ed Sauna ZE, Kimchi-Sarfaty C. Cham: Springer International Publishing
124. Zhao F, Yu C-H, Liu Y. 2017. Codon usage regulates protein structure and function by affecting translation elongation speed in Drosophila cells. *Nucleic Acids Res.* 45(14):8484–92 [PubMed: 28582582]
125. Zhou M, Guo J, Cha J, Chae M, Chen S, et al. 2013. Non-optimal codon usage affects expression, structure and function of clock protein FRQ. *Nature.* 495:111–15 [PubMed: 23417067] --- codons can significantly affect folding and function of a protein lacking a well-defined native structure
126. Zhou M, Wang T, Fu J, Xiao G, Liu Y. 2015. Nonoptimal codon usage influences protein structure in intrinsically disordered regions. *Mol Microbiol.* 97(5):974–87 [PubMed: 26032251]
127. Zhou Z, Dang Y, Zhou M, Li L, Yu CH, et al. 2016. Codon usage is an important determinant of gene expression levels largely through its effects on transcription. *PNAS.* 113(41):E6117–25 [PubMed: 27671647]

SUMMARY POINTS

- For many proteins, the rate of folding is set by the rate of translation elongation.
- Synonymous codon substitutions can change elongation rate ten-fold without constraining the encoded amino acid sequence.
- Increasing elongation rate is not always “optimal” for supporting proper protein folding.
- Synonymous codon substitutions can act in concert with non-synonymous substitutions to amplify or mitigate effects on protein folding.
- In addition to direct effects on co-translational folding, synonymous substitutions can also disrupt entry of proteins into the secretory pathway.
- It is unfortunately easy to misattribute the effects of synonymous codon substitutions, because these effects can be subtle and affect diverse aspects of functional protein production.
- The effects of synonymous codons on co-translational folding and other aspects of protein production may influence evolutionary pathways.

FUTURE ISSUES

- Both protein folding intermediates and translation intermediates are fleeting and therefore hard to study, which hampers the advancement of our understanding of the impact of codon substitutions on protein folding.
- The lack of a model to predict elongation rate accurately and quantitatively for any mRNA sequence of interest has led to a reliance on imperfect proxies, such as equating rare codons and/or low tRNA gene copy number with slow elongation rate.
- Historically, most genome-wide association studies have disregarded disease-associated synonymous substitutions, which has hindered our appreciation of the contributions of synonymous mutations to disease.

SIDEBAR 1:**Why are commonly-used codons typically associated with rapid elongation rate – and “codon optimality”?**

Currently, we lack experimental tools capable of accurately measuring absolute elongation rates of individual codons in cells (although recent advancements (77) are bringing us closer to this goal). As a result of this limitation, the field relies on imperfect proxies of relative elongation rate. One of the oldest and most widely used proxies for elongation rate is codon usage bias (see also Sidebar 2). It has long been known that the coding sequences of highly abundant proteins are enriched in a subset of synonymous codons that are also positively correlated with high isoacceptor tRNA abundance, genome-wide abundance and more rapid translation elongation (46, 65, 67, 89, 122). Discovery of these so-called “optimal” codons (53) was crucial for the success of heterologous gene expression experiments, where organism-specific differences in codon optimality can introduce sub-optimal codons when transplanting a gene from one organism to another. Substituting these sub-optimal codons to optimal ones for the expression host can greatly enhance protein abundance, but may also lead to protein misfolding, aggregation and/or degradation. An alternative strategy, codon harmonization (6, 84, 118), retains codon usage patterns and can improve protein folding yield, despite lower overall protein accumulation (111).

SIDEBAR 2:**Levels of codon usage bias.**

In all organisms, there are at least three distinct levels of synonymous codon usage bias. Genome-level codon usage bias means that each organism uses some synonymous codons more than others, genome-wide. This bias is what leads to different genomic %GC content. In contrast, gene-level codon usage bias means that some genes in a genome are enriched or depleted in certain synonymous codons, relative to the genomic average. One of the earliest and most striking correlations detected for gene-level bias is that highly expressed genes are enriched in a subset of codons (46). In vertebrates, large groups of neighboring genes called isochores have distinct codon usage frequencies from genes that lie outside the isochores (36). Intra-genic codon usage bias refers to a non-random distribution of codon usage within a single coding sequence. One of the most striking examples of intragenic codon usage bias is the 5' end of coding sequences, which are enriched in rarely used (from a genomic perspective) codons that tend to destabilize mRNA structure, leading to more efficient translation initiation (20, 71).

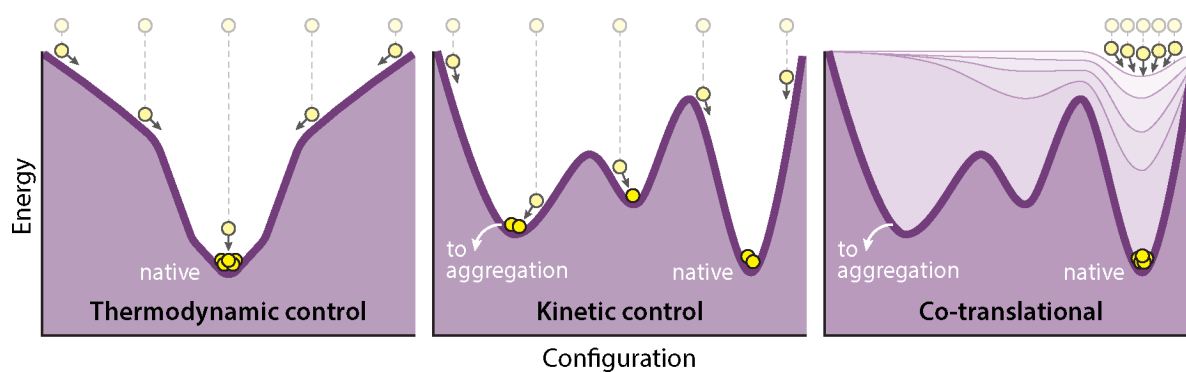


Figure 1.

Energy landscapes for (a) a simple model protein that folds reversibly in the Anfinsen experiment (see text), (b) a typical protein prone to aggregation, and (c) an example of how co-translational folding could help promote folding and avoid aggregation for a typical protein. Diagrams inspired by (11, 34).

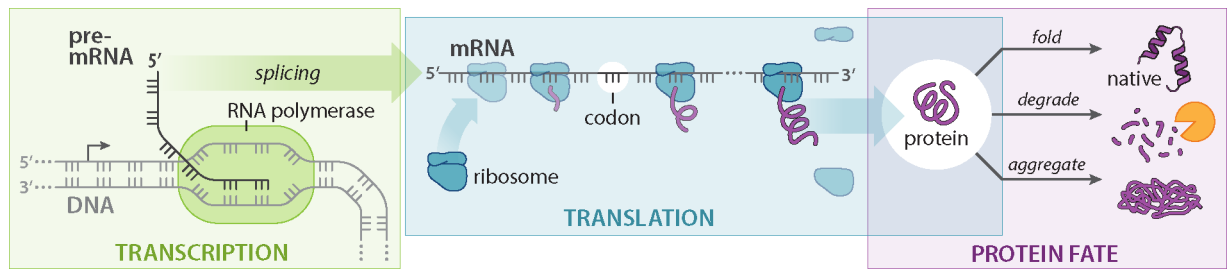


Figure 2.

This review focuses on how synonymous codon-induced effects on elongation rate (*blue*) can influence the success of protein folding (*purple*) and membrane targeting (not shown). This figure serves as a reminder that codon usage can affect many other processes, including transcription level and splicing (*green*), translation initiation and elongation rate (blue), translational fidelity (not shown) and mRNA degradation (not shown).

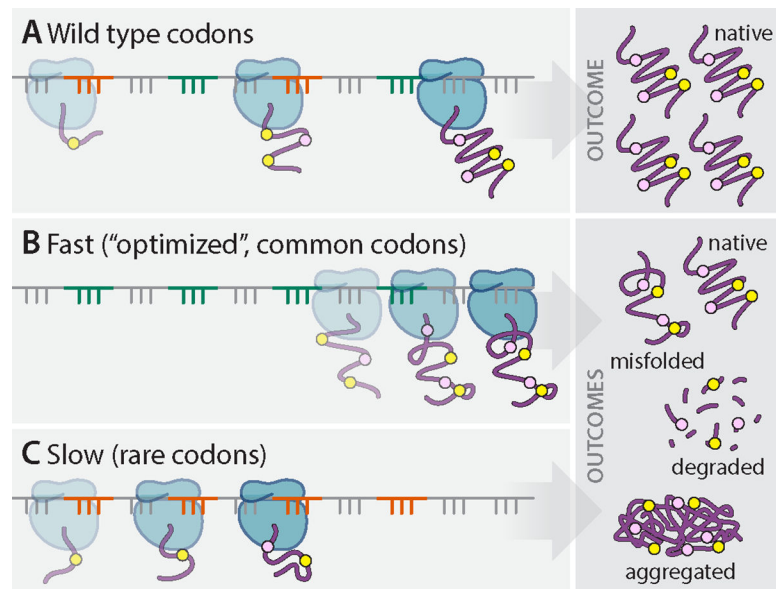


Figure 3.

Wild type mRNA transcripts (*a*) typically contain both rare and common codons for efficient translation and functional protein production. Synonymous codon substitutions to codons that are more optimized (common) (*b*) or de-optimized (rare) (*c*) can modify protein folding pathways, leading to misfolding, aggregation and/or degradation.

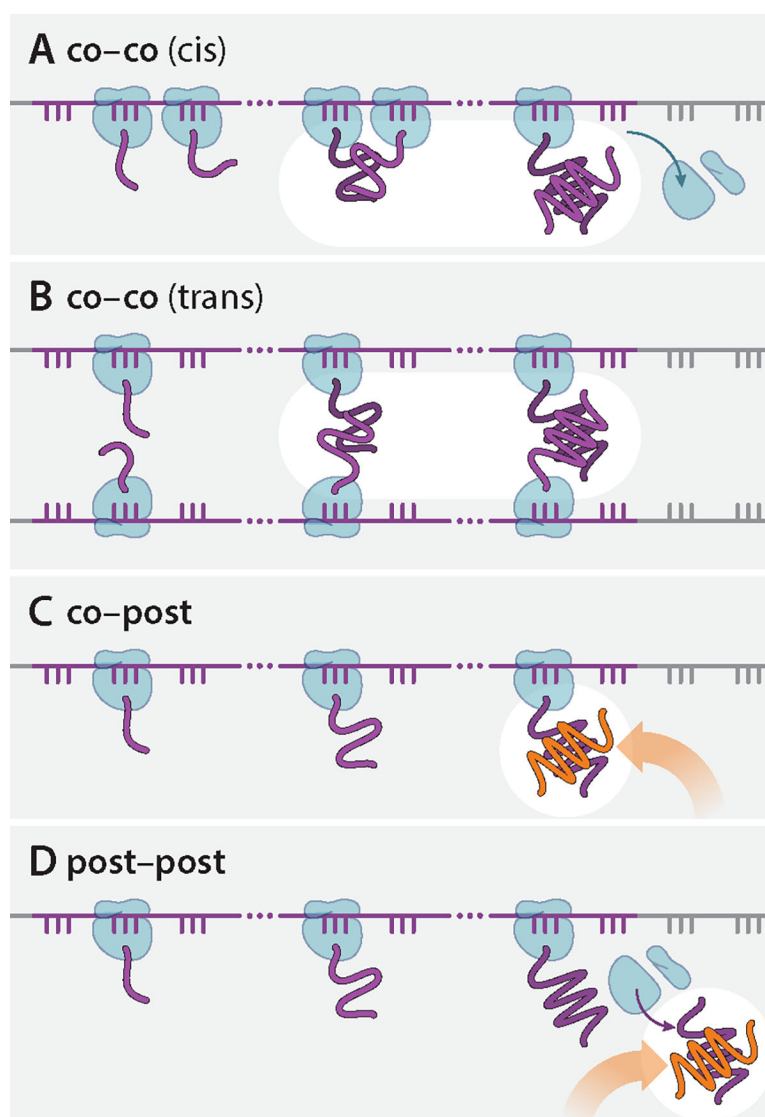


Figure 4. The four potential assembly mechanisms for multimeric proteins. Of note, co-co cis assembly is specific to homomeric proteins. This figure was inspired by Bertolini et al., 2021 (18).

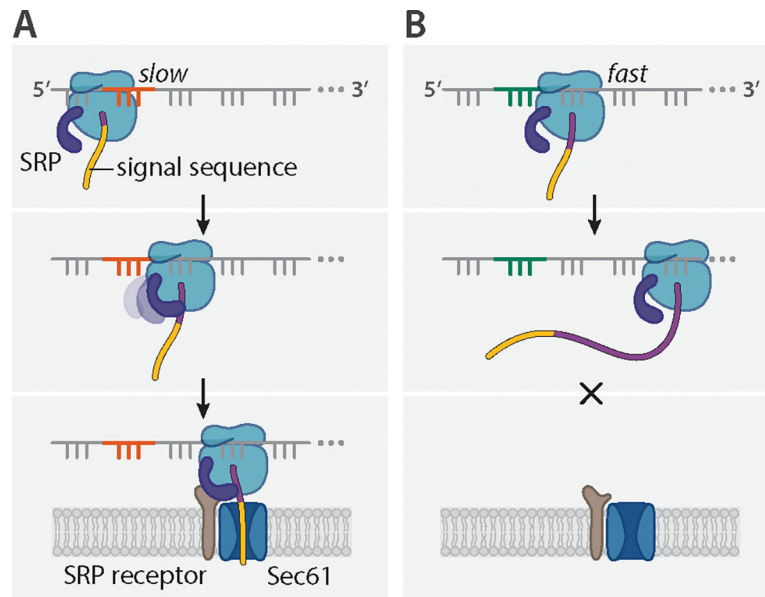


Figure 5. Codon-mediated SRP binding. (a) Non-optimal codons (*red*) tend to be translated slowly, which can facilitate SRP (*navy*) binding to the signal sequence (*yellow*) of the RNC, pausing translation elongation until the ribosome docks with the SRP receptor (*brown*) and Sec61 translocon (*blue*) at the ER membrane. (b) In contrast, rapid elongation by optimal codons (*green*) means the signal sequence might slip past SRP without binding. Without SRP binding, elongation will not pause and the RNC will not properly engage with the translocon at the ER membrane.

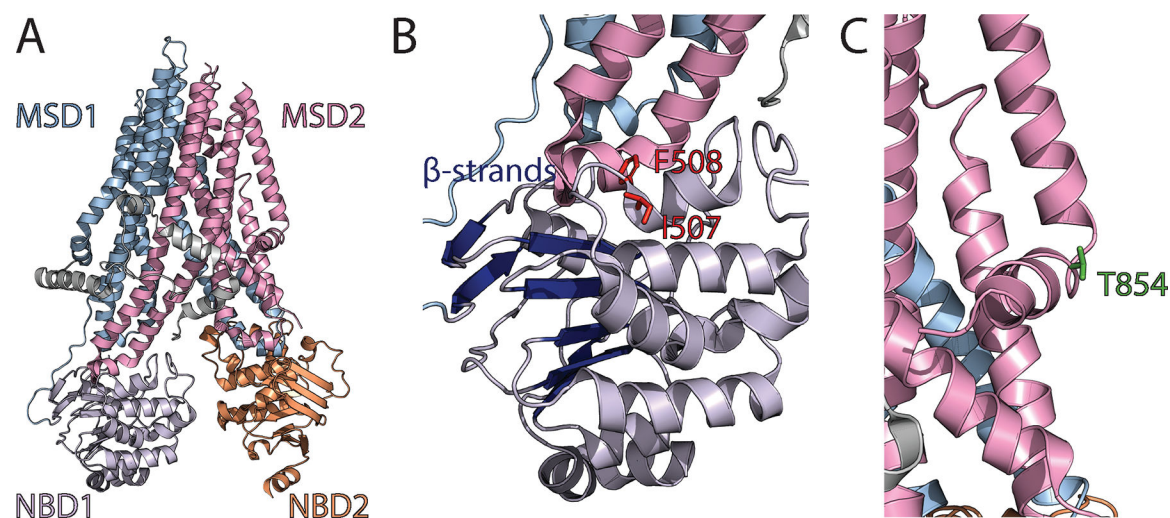


Figure 6.

Sites in CFTR affected by codon usage. (a) A ribbon diagram of CFTR (PDB:5uak) depicts membrane-spanning domain 1 (MSD1, *blue*), NBD1 (*purple*), MSD2 (*pink*), and NBD2 (*orange*). (b) A view of NBD1 (*purple*) depicting I507 and F508 (*red*) and the β -strands in NBD1 (*navy*). (c) A view of MSD2 (*pink*) depicting T854 (*green*).